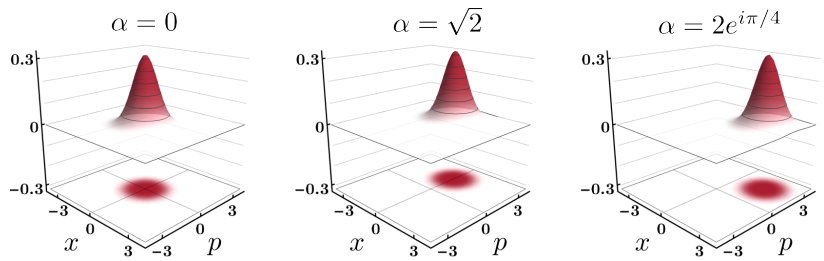

Figure 3.4: Wigner functions for coherent states with (x_α, p_α) equal to $(0,0)$, $(2,0)$, and $(2,2)$, from left to right.

The P-function

The Glauber-Sudarshan P-function (or P-representation) gives another quasiprobability distribution on phase space. Any state $\hat{\rho}$ can be expressed on a diagonal form in terms of coherent states as

$$\hat{\rho} = \int P_\rho(\alpha)|\alpha\rangle\langle\alpha|\, d^2\alpha, \tag{3.87}$$

where the integral is over the complex plane and P_ρ is the P-function. This is possible because the coherent states are not orthogonal and form an overcomplete set. P_ρ is real because $\hat{\rho}$ is Hermitian and normalised because $\int P_\rho(\alpha)\, d^2\alpha = \mathrm{Tr}(\hat{\rho}) = 1$. Like the Wigner function, however, P_ρ can become negative or highly singular for some states and thus is not a true probability distribution.

The Q-function

A third phase-space distribution, which is sometimes useful, is the Husimi Q-function, defined in terms of the expectation value of $\hat{\rho}$ with respect to a coherent state

$$Q_\rho(\alpha) = \frac{1}{\pi}\langle\alpha|\hat{\rho}|\alpha\rangle. \tag{3.88}$$

The Q-function is real, positive, and bounded $0 \leq Q_\rho(\alpha) \leq \frac{1}{\pi}$, and it is normalised because of Eq. (3.74). It is thus a proper probability distribution over the complex plane. Note that, because of (3.74), the operators

$$\hat{E}_\alpha = \frac{1}{\pi}|\alpha\rangle\langle\alpha|, \tag{3.89}$$

form a POVM (with continuous outcome). $Q_\rho(\alpha) = \mathrm{Tr}[\hat{E}_\alpha\hat{\rho}]$ is the probability for obtaining outcome α when performing the corresponding measurement on a system in state $\hat{\rho}$. The measurement is not projective because different coherent states are not orthogonal. Such a measurement can be realised in optical systems by heterodyne detection, as discussed in Sec. 3.2.5.

The characteristic function

The three quasiprobability distributions above (Wigner, P- and Q-functions) are in fact closely related and can all be expressed in terms a so-called charac-

teristic function. The p-ordered characteristic function is defined to be

$$\chi_{\rho,p}(\alpha) = \mathrm{Tr}[\hat{\rho}\hat{D}(\alpha)]e^{\frac{1}{2}p|\alpha|^2} = \mathrm{Tr}[\hat{\rho}e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}]e^{\frac{1}{2}p|\alpha|^2}, \tag{3.90}$$

where the parameter p can take three values, $1, 0$, and -1. Each value corresponds to a particular ordering of creation and annihilation parameters in the expression. Specifically, it can be shown that

$$\chi_{\rho,1}(\alpha) = \mathrm{Tr}[\hat{\rho}\, e^{\alpha\hat{a}^\dagger} e^{-\alpha^*\hat{a}}] \qquad \text{(normal)}, \tag{3.91}$$

$$\chi_{\rho,0}(\alpha) = \mathrm{Tr}[\hat{\rho}\, e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}] \qquad \text{(symmetric)}, \tag{3.92}$$

$$\chi_{\rho,-1}(\alpha) = \mathrm{Tr}[\hat{\rho}\, e^{-\alpha^*\hat{a}} e^{\alpha\hat{a}^\dagger}] \qquad \text{(anti-normal)}, \tag{3.93}$$

such that $p = 1$ corresponds to normal order ($\hat{a}^\dagger$ before $\hat{a}$), $p = -1$ to anti-normal order ($\hat{a}$ before $\hat{a}^\dagger$), and $p = 0$ to symmetric order (average of all possible orders). The quasiprobability distributions are given by a two-dimensional Fourier transform of the characteristic function

$$\frac{1}{\pi^2}\int \chi_{\rho,p}(\beta)e^{\alpha\beta^* - \alpha^*\beta}d^2\beta = \begin{cases} P_\rho(\alpha) & \text{for } p = 1 \\ W_\rho(\alpha) & \text{for } p = 0 \\ Q_\rho(\alpha) & \text{for } p = -1 \end{cases}, \tag{3.94}$$

where the integral is over the complex plane and $W(\alpha)$ should be interpreted as $W(x, y)$ with $\alpha = (x + iy)/\sqrt{2}$.

Characteristic functions and quasiprobability distributions are powerful tools with many uses in theoretical quantum optics and beyond. This is outside the scope of these notes, however, and we will not delve further into the details. For more information see e.g. the books by Barnett and Radmore [1] or Gerry and Knight [5]. Here, we will only be needing the Wigner function.

3.1.8 Squeezed states

So far, we have discussed Fock states and coherent states. There are of course many other states of the harmonic oscillator which are interesting and useful for quantum technology. Among these, the so-called *squeezed states* are of particular importance, because they are relatively easy to generate and at the same time enable superclassical performance in many applications.

In general, squeezing can be defined in any system (not just harmonic oscillators). Given two observables $\hat{A}$ and $\hat{B}$, the uncertainty relation (2.24) states that $\langle\Delta^2\hat{A}\rangle\langle\Delta^2\hat{B}\rangle \geq \frac{1}{4}|\langle[\hat{A}, \hat{B}]\rangle|^2$ in any state. A state is said to be squeezed if

$$\langle\Delta^2\hat{A}\rangle < \frac{1}{2}|\langle[\hat{A}, \hat{B}]\rangle| \qquad \text{or} \qquad \langle\Delta^2\hat{B}\rangle < \frac{1}{2}|\langle[\hat{A}, \hat{B}]\rangle|, \tag{3.95}$$

i.e. one of the observables is contributing less uncertainty than an equal share of the minimum joint uncertainty. When the system consists of harmonic oscillators (one or several) and $\hat{A}$ and $\hat{B}$ are chosen to be (combinations of) quadratures, this is referred to as *quadrature squeezing*. This is the only kind of squeezing we will be concerned with here.